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Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P08

Set 8

Time: 25 minutes | Paper Reference **PMAT2/N08**

Mathematics

Level 2
Section A (Non–Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
16 /16

My signature confirms that I will not discuss the content of the test with anyone.

Signature: ~~Master Functional Skills (MFS)~~

Instructions

- Use **black** ink or ball-point pen.
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- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used.**
- Take the value of π to be 3.14

Information

- The total mark for this section is 16.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ▶

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1 Eve buys lunch on her way to work.
She sees this meal deal.

Meal deal

Buy any sandwich
snack and drink
for £3.99

Eve chooses these items.

Chicken sandwich	£2.85
Fruit snack pot	79p
Coffee	£1.59

Work out how much money Eve will save using the meal deal.
You **must** show your working.

(3)

Step 1: Calculate total cost of items bought separately

(carries) 2 2

£ 2.85 (chicken sandwich)

\+ 0.79 (fruit snack pot: 79p = £0.79)

1.59 (coffee)

£ 5.23 [M1] Total separate cost = £5.23

Step 2: Subtract meal deal price (£3.99) to find savings

(borrowing) 4 11 13

£ ~~5~~.~~2~~3 (separate items)

− 3.99 (meal deal price)

[M1] Subtraction with borrowing shown

£ 1.24 [A1] Eve saves £1.24 using the deal

£ 1.24

(Total for Question 1 is 3 marks)

2

Work out 74.88 ÷ 1.2
You **must** show your working.

(3)

Step 1: Make divisor a whole number (multiply both numbers by 10)

$74.88 \div 1.2 = (74.88 \times 10) \div (1.2 \times 10)$

$= 748.8 \div 12$ *[M1] Equivalent division by integer*

Step 2: Short division (bus stop method)

$$12 \overline{)\, 7^{7}\,4^{2}\,8.^{4}\,8} \quad = \quad 0\;6\;2.\;4$$

- *12 into 7 = 0 r 7 -> 12 into 74 = 6 r 2*
- *12 into 28 = 2 r 4 (2 × 12 = 24, 28–24=4)*
- *12 into 48 = 4 r 0 (4 × 12 = 48 exact)*
- *Decimal point aligns directly above*

Result = 62.4

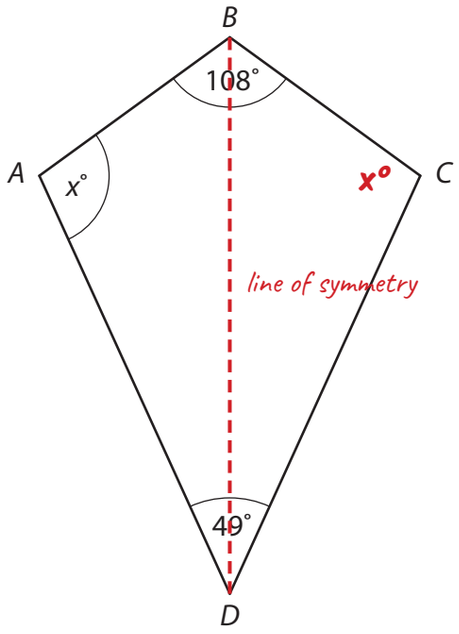
[M1] Digits 624 identified
[A1] Accurate final answer: 62.4

62.4

(Total for Question 2 is 3 marks)

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3 *ABCD* is a 4-sided shape.
BD is a line of symmetry of the shape.



(a) Work out the value of *x*.

(3)

Step 1: Sum of angles in a quadrilateral = 360°

Sum of known angles: 108° + 49° = 157° [M1] Sum = 157°

Remaining angle sum = 360° − 157° = 203°

BD is a line of symmetry, so Angle BAD = Angle BCD = x°

2x = 203° => x = 203° ÷ 2 [M1] 203 ÷ 2

x = 101.5° [A1] Accurate angle: 101.5° (200÷2=100, 3÷2=1.5)

101.5

(b) Use estimation to check your answer.

(1)

Round angles to nearest 10°: 108° ≈ 110°, 49° ≈ 50°

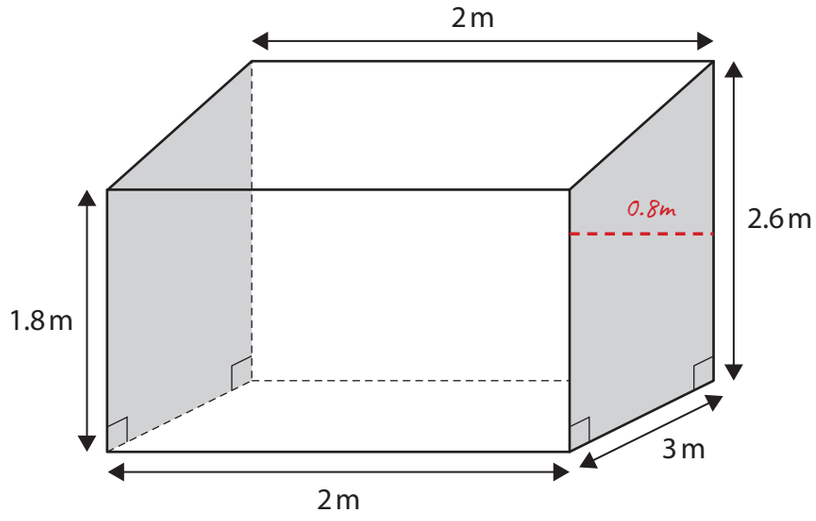
Check: (360° − 110° − 50°) ÷ 2 = 200° ÷ 2 = 100°

100° is very close to 101.5° ☑ (Sensible check confirmed) [B1] Valid check

(Total for Question 3 is 4 marks)

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4 Max is building a new shed.
He has drawn this sketch.



Each shaded wall of the shed is in the shape of a trapezium.
The other two walls are rectangular.

Max wants to cover the four walls with wood cladding.
He will not use any cladding for the floor or the roof.

Max will buy the cladding in packs.
Each pack of cladding covers 1.1 m^2

The cladding can be cut and joined.

Max knows that the two rectangular walls need 8.8 m^2 of cladding in total.

Work out how many packs of cladding Max needs to cover the four walls.

(6)

Step 1: Missing height of front trapezium wall

Height = 2.6 m − 1.8 m = 0.8 m *[P1] Height = 0.8m*

Step 2: Area of ONE shaded trapezium wall (split into 2 shapes):

- **Rectangle = 3 m × 1.8 m = 5.4 m^2** *[P1] Rectangle = 5.4 m^2*
- **Triangle = 0.5 × 3 m × 0.8 m = 1.2 m^2**
- **Area of 1 wall = 5.4 + 1.2 = 6.6 m^2** *[P1] 1 wall = 6.6 m^2*

(Or formula: Area = ½(1.8 + 2.6) × 3 = 2.2 × 3 = 6.6 m^2)

-> See next page for total area of 4 walls & pack calculation...

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Step 3: Total area to clad across all four walls

- 2 Shaded trapezium walls = 2 × 6.6 m² = 13.2 m²
- 2 Rectangular walls = 8.8 m² (stated in question)

Total cladding area required:

13.2 m² + 8.8 m² = 22.0 m² [P1] Total area = 22 m²

Step 4: Number of cladding packs (each pack covers 1.1 m²)

Packs needed = Total area ÷ 1.1 m²

= 22 ÷ 1.1 (multiply by 10: 220 ÷ 11) [P1] 22 ÷ 1.1

= 220 ÷ 11 = 20 packs [A1] Accurate packs: 20

☒ Full Verification Summary:

- 2 Trapezium walls: 2 × 6.6 m² = 13.2 m²
- 2 Rectangular walls: 8.8 m²
- Total Area: 13.2 + 8.8 = 22 m² => 20 packs × 1.1 m² = 22 m² (exact match ☒)

20

(Total for Question 4 is 6 marks)

TOTAL FOR SECTION A IS 16 MARKS

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Set 8

Time: 1 hour 30 minutes | Paper Reference **PMAT2/C08**

Mathematics

Level 2
Section B (Calculator)



You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
48 /48

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Signature: ~~Master Functional Skills (MFS)~~

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- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used.**
- If your calculator does not have a π button take the value of π to be 3.14

Information

- The total mark for this section is 48.
- The total mark for this paper is 64.
- The marks for **each** question are shown in brackets
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Advice

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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Mandy spins a fair spinner twice.

The colours on the spinner are red, blue, green, orange and pink.
The table shows all the possible outcomes.

		Second spin				
		Red	**Blue**	**Green**	**Orange**	**Pink**
First spin	**Red**	R, R	R, B	R, G	R, O	R, P
	Blue	B, R	B, B	B, G	B, O	B, P
	Green	G, R	G, B	G, G	G, O	G, P
	Orange	O, R	O, B	O, G	O, O	O, P
	Pink	P, R	P, B	P, G	P, O	P, P

(a) What is the probability of getting the same colour on both spins?

(1)

Total possible outcomes = 5 × 5 = 25

Same colour on both spins: (R,R), (B,B), (G,G), (O,O), (P,P) = 5 outcomes

P(same colour) = 5 / 25 (= 1/5 = 0.2) *[B1] 5/25 or 1/5*

5/25

(b) What is the probability of getting at least one blue in the two spins?

(2)

Outcomes with AT LEAST ONE Blue in the two spins:

- *1st spin Blue: (B,R), (B,B), (B,G), (B,O), (B,P) -> 5 outcomes*
- *2nd spin Blue: (R,B), (G,B), (O,B), (P,B) -> 4 other outcomes*

Total outcomes with at least one Blue = 5 + 4 = 9 outcomes *[M1] 9 outcomes identified*

P(at least one Blue) = 9 / 25 (= 0.36 = 36%) *[A1] Accurate probability: 9/25*

9/25

(Total for Question 1 is 3 marks)

P 6 8 4 7 1 A 0 2 2 0

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2 Habiba wants to make pastry by mixing butter and flour.
She needs to mix the butter and the flour in the ratio 4 : 9

Habiba wants to make 650 grams of pastry.
She has 425 grams of flour.

Does Habiba have enough flour to make 650 grams of pastry?
Show why you think this.

(3)

Step 1: Calculate total ratio parts for pastry (butter : flour = 4 : 9)

Total parts = 4 + 9 = 13 parts [M1] Ratio parts: 4 + 9 = 13

Step 2: Find weight of 1 part for 650 g of pastry

1 part = 650 g ÷ 13 = 50 g (Since 13 × 5 = 65, so 650 ÷ 13 = 50)

Step 3: Calculate flour required (9 parts)

Flour needed = 9 parts × 50 g = 450 g [A1] Full process: 9 × 50 = 450 g

Step 4: Compare required flour with available flour (425 g)

Habiba has 425 g of flour, but needs 450 g.

Shortage = 450 g − 425 g = 25 g of flour short

Conclusion: No, Habiba does NOT have enough flour. [C1] Final conclusion: No with 450 g

No, needs 450g (25g short)

(Total for Question 2 is 3 marks)

3 Habiba also wants to make cupcakes.
She has this list of ingredients.

Cupcakes
makes 12

2 cups of flour	1 egg
3 tbsp butter	10.5 fl oz milk
$\frac{3}{4}$ cup sugar	3 apples
pinch of cinnamon	

Habiba wants to make 30 cupcakes.

1 fl oz = 28.413 ml

How much milk does Habiba need to make 30 cupcakes?
Give your answer to the nearest ml.

(4)

Step 1: Scaling factor to make 30 cupcakes (recipe makes 12)

Scale factor = 30 ÷ 12 = 2.5 *[M1] Proportion scale = 2.5*

Step 2: Calculate milk needed for 30 cupcakes in fluid ounces (fl oz)

Milk needed = 10.5 fl oz × 2.5 = 26.25 fl oz *[M1] 10.5 × 2.5 = 26.25 fl oz*

Step 3: Convert fl oz to millilitres (ml) using 1 fl oz = 28.413 ml

Milk in ml = 26.25 × 28.413 = 745.84125 ml *[M1] 26.25 × 28.413 = 745.84...*

Step 4: Round to nearest ml (as requested in question)

745.84125 ml ≈ 746 ml (since tenths digit 8 ≥ 5, round UP) *[A1] Accurate integer: 746 ml*

746 ml

(Total for Question 3 is 4 marks)

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4 Callum is a plumber.
He spends money on eight different adverts.

Callum has this information about the money he spent on each advert and the number of new customers from seeing that advert.

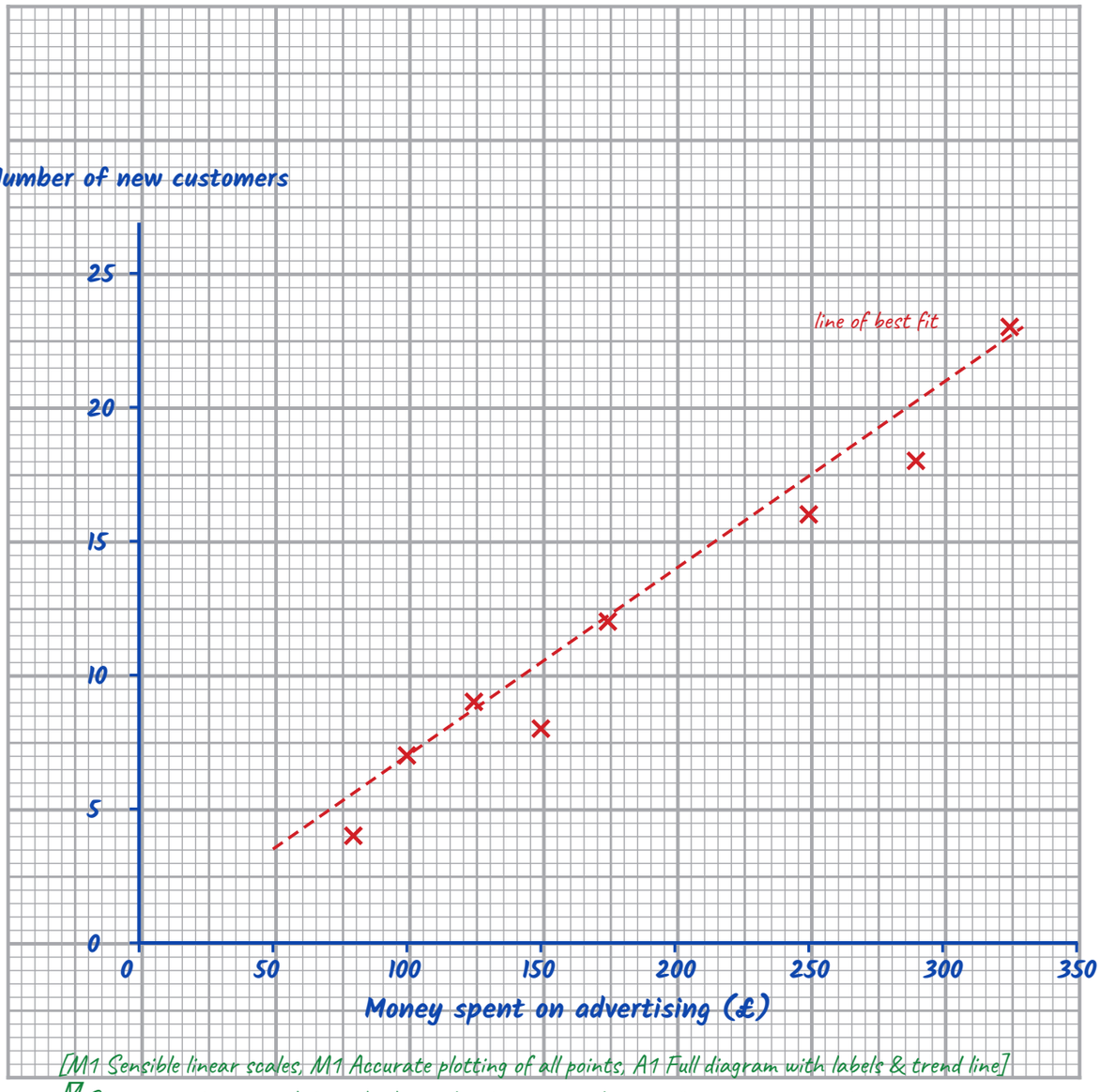
Money spent (£)	100	175	80	250	150	325	125	290
Number of new customers	7	12	4	16	8	23	9	18

Callum wants to draw a diagram to see if there is a relationship between the money he spent on each advert and the number of new customers from seeing that advert.

Draw a suitable diagram for Callum.

(3)

Scatter Diagram: Advertising Cost vs Customers



[M1 Sensible linear scales, M1 Accurate plotting of all points, A1 Full diagram with labels & trend line]

Strong positive correlation: higher advertising spend generates more new customers.

(Total for Question 4 is 3 marks)

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5 Misha rows in a team of 8 people.

She wants to compare the mean BMI of the people in the rowing team with the mean BMI of women in the UK.

Misha uses this formula to calculate her BMI.

$$\text{BMI} = \frac{W}{H^2}$$

W is weight (kg)
H is height (metres)

Misha has a weight of 73.3 kg and a height of 67 inches.

1 inch = 2.54 cm

The table shows some information about the BMIs of the people in the rowing team.

Louise	Katia	Leila	Lisa	Gemma	Mel	Steph	Misha
24.8	27.1	25.7	28.3	26.4	25.2	27.6	25.31 .....

The mean BMI of women in the UK is 26.9

Compare the mean BMI of the people in the rowing team with the mean BMI of women in the UK.

(5)

Step 1: Convert Misha's height from inches to metres (1 inch = 2.54 cm)

Height in cm = 67 inches × 2.54 cm = 170.18 cm

Height in metres = 170.18 ÷ 100 = 1.7018 m

[M1] Height in metres: 1.7018m

Step 2: Calculate Misha's BMI using formula BMI = $W \div H^2$

$H^2 = (1.7018)^2 = 2.896123...\ m^2$ [M1] Begins BMI formula: H^2

Misha's BMI = $73.3 \div 2.896123... = 25.3096... \approx 25.31$ [A1] Full BMI process = 25.31

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Step 3: Calculate the mean BMI of all 8 rowing team members

Total BMI = 24.8 + 27.1 + 25.7 + 28.3 + 26.4 + 25.2 + 27.6 + 25.31

= 210.41

Mean BMI = 210.41 ÷ 8 = 26.30125 ≈ 26.30

[M1] Team mean BMI = 26.3

Step 4: Compare team mean BMI with UK women's mean BMI (26.9)

- Rowing team mean BMI = 26.3
- UK women mean BMI = 26.9
- 26.3 < 26.9 (Team mean is lower by 0.6)

Final Comparison Statement:

The mean BMI for Misha's rowing team is 26.3, which is lower than the mean BMI of 26.9 for women in the UK.

[A1] Valid comparison with accurate figures (26.3 < 26.9)

(Total for Question 5 is 5 marks)

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6

(a) Write three million, six hundred and seventy five thousand in figures.

(1)

Place value breakdown: 3 millions + 675 thousands = 3,675,000

Three million, six hundred and seventy five thousand = 3,675,000

[B1] Correct value

3,675,000

(b) Write 760 grams as a fraction of 1 kilogram.
Give your answer in its simplest form.

(2)

Step 1: Convert 1 kilogram to grams (1 kg = 1000 g)

760 grams as a fraction of 1000 grams = 760 / 1000

[M1] Fraction 760/1000

Step 2: Simplify to lowest terms (divide by common factors)

760 / 1000 = 76 / 100 (divide top & bottom by 10)

= 38 / 50 (divide top & bottom by 2)

= 19 / 25 (divide top & bottom by 2)

[A1] Simplified form 19/25

$$\frac{19}{25}$$

(Total for Question 6 is 3 marks)

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7 Nyam looks at the screen time report on his phone.

This week his average screen time per day is 28% greater than his average screen time per day last week.

This week Nyam's average screen time per day is 3 hours and 44 minutes.

Work out Nyam's average screen time per day for last week.

(4)

Step 1: Convert this week's screen time to total minutes

3 hours 44 minutes = (3 × 60) + 44 = 180 + 44 = 224 minutes

[M1] Conversion: 224 mins

Step 2: Express this week as percentage of last week (original = 100%)

This week = 100% + 28% = 128%

128% of last week = 224 minutes

[M1] Reverse percentage: 128% = 224

Step 3: Calculate 100% (last week's screen time)

1% = 224 ÷ 128 = 1.75 minutes [M1] 224 ÷ 1.28 = 175 mins

100% = 1.75 × 100 = 175 minutes (or 224 ÷ 1.28 = 175 mins)

Step 4: Convert 175 minutes into hours and minutes

175 ÷ 60 = 2 hours remainder 55 minutes [A1] 2 hours 55 minutes (or 175 mins)

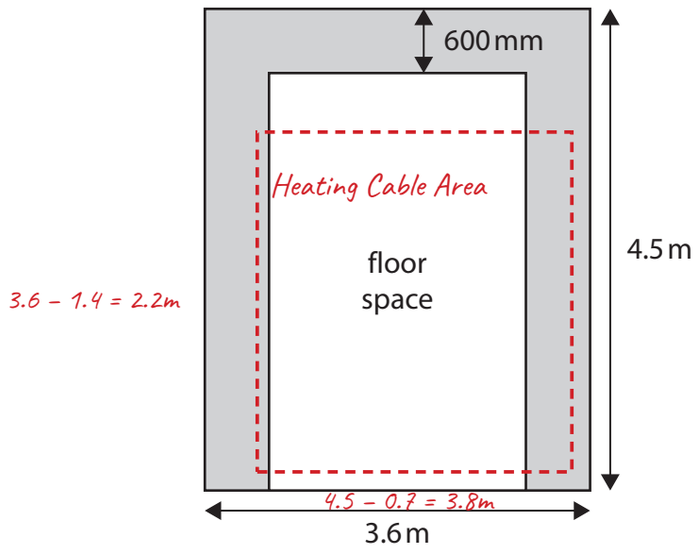
= 2 hours 55 minutes (2 × 60 = 120 mins, 175 − 120 = 55 mins)

2 hours 55 mins

(Total for Question 7 is 4 marks)

8 Frank wants to fit underfloor heating in his kitchen.
The kitchen is in the shape of a rectangle.

He has this sketch.



The shaded space shows the part of the floor covered by cabinet bases.
Each cabinet has a depth of 600 mm.

Frank will cover part of the floor space with heating cable.

Frank needs to leave a gap of 100 mm between the base of the cabinets and the part of the floor space he will cover.

He needs to buy a cable pack.

The table shows information about the different sizes of heating cable packs and the maximum floor area each pack can cover.

Cable pack size	Maximum floor area (m^2)	Cost (£)
small	4.2	109.99
medium	6.0	164.99
large	8.4	179.99
extra large	10.6	199.99

Frank wants to spend as little as possible.

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(a) Work out the cost of the cable pack Frank needs to buy.
You **must** show your working.

(5)

Step 1: Convert mm to metres and calculate border margins

Cabinet depth = 600 mm = 0.6 m, Gap required = 100 mm = 0.1 m

Total margin from walls with cabinets = 0.6 m + 0.1 m = 0.7 m *[M1] Consistent units: 0.7 m*

Step 2: Find dimensions of heated floor area

Width = 3.6 m − 0.7 m − 0.7 m = 2.2 m *[M1] Width = 2.2 m* **(cabinets on both sides)**

Length = 4.5 m − 0.7 m = 3.8 m *[M1] Length = 3.8 m* **(cabinets on one side)**

Step 3: Calculate required heating cable area

Area = 2.2 m × 3.8 m = 8.36 m² *[M1] Heating area = 8.36 m²*

Step 4: Select cheapest pack that covers at least 8.36 m²

- *Small: max 4.2 m² (Too small), Medium: max 6.0 m² (Too small)*
- **Large: max 8.4 m² covers 8.36 m² (8.36 ≤ 8.4 -> Suitable & Cheapest)**

Cost of Large pack = £179.99 *[A1] Large pack: £179.99*

£ **179.99**

(b) Use a reverse calculation to show a check of your answer.

(1)

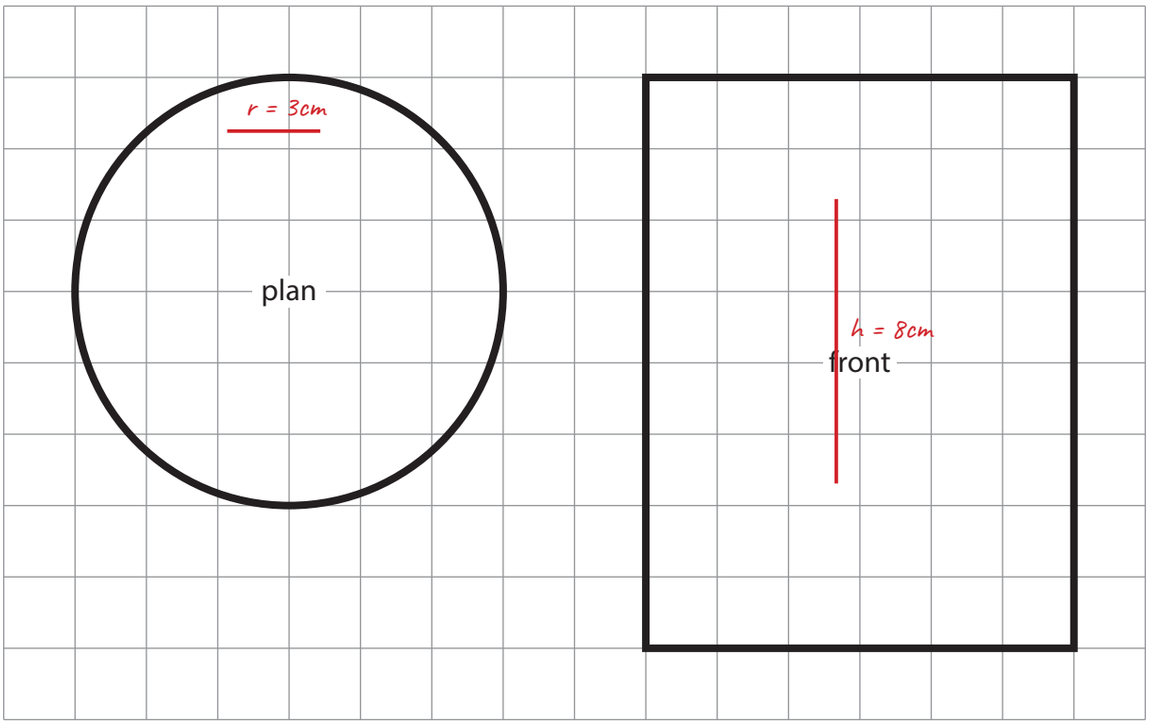
Reverse calculation checking area multiplication:

8.36 m² ÷ 3.8 m = 2.2 m ☐ *[B1] Valid reverse check*

(Or: 8.36 m² ÷ 2.2 m = 3.8 m ☒)

(Total for Question 8 is 6 marks)

9 Sammi has designed a sweet tin in the shape of a cylinder.
The plan and the front elevation of the tin are shown on the centimetre grid below.



Find the volume of the sweet tin.

(3)

Step 1: Read dimensions from centimetre grid (1 square = 1 cm)

- **Plan (circle): Diameter = 6 cm => Radius, r = 3 cm**
- **Front elevation: Height, h = 8 cm** [M1] Reads r = 3 cm, h = 8 cm

Step 2: Calculate cylinder volume using formula: $V = \pi r^2 h$

Circular Base Area = $\pi \times r^2 = 3.14 \times 3^2 = 3.14 \times 9 = 28.26$ cm²

Volume = Base Area × h = 28.26 cm² × 8 cm [M1] Full volume process: $\pi \times 3^2 \times 8$

= 226.08 cm³ (or 226 cm³ to 3 s.f.) [A1] Accurate volume: 226.08 cm³

226.08 cm³

(Total for Question 9 is 3 marks)

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10

(a) Write $\frac{15}{7}$ as a mixed number.

(1)

Divide numerator by denominator (15 ÷ 7):

15 ÷ 7 = 2 with remainder 1

Whole number = 2, Numerator = 1, Denominator = 7

15 / 7 = 2 1/7

[B1] Mixed number: 2 1/7

$2\frac{1}{7}$

Here is a list of numbers.

−3 6 −5 4 4 0

(b) Find the median.

(2)

Step 1: Put the 6 numbers in order from smallest to largest:

−5, −3, 0, 4, 4, 6 (middle two values)

Step 2: Find the midpoint (mean) of the middle two values (0 and 4)

[M1] Ordered list & middle values

Median = (0 + 4) ÷ 2 = 4 ÷ 2 = 2

[A1] Median = 2

2

(Total for Question 10 is 3 marks)

11

(a) Write 28% as a decimal.

(1)

Convert percentage to decimal by dividing by 100:

28% = 28 ÷ 100 = 0.28 *[B1] Accurate decimal: 0.28*

0.28

Andrew pays 38.90 euros for a meal in Spain using his British bank card.

The British bank charges a fee for changing currency.
The fee is 2.75% of the amount he pays for the meal.

£1 = 1.127 euros

(b) Work out the total amount Andrew pays for the meal and the fee.
Give your answer in pounds.

(4)

Step 1: Calculate the bank currency exchange fee in euros

Fee = 2.75% of €38.90 = (2.75 ÷ 100) × 38.90

= 0.0275 × 38.90 = €1.06975 *[M1] Percentage fee: €1.06975*

Step 2: Total amount paid in euros (meal + fee)

Total in euros = €38.90 + €1.06975 = €39.96975 *[M1] Total in euros: €39.97*

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Step 3: Convert total amount from euros to British pounds (£1 = 1.127 euros)

Total in pounds (£) = Total euros ÷ 1.127

= 39.96975 ÷ 1.127 [M1] Currency conversion: divide by 1.127

= £35.46561668...

Step 4: Round to 2 decimal places for pounds and pence

£35.4656... ≈ £35.47 (since 3rd decimal digit 5 ≥ 5, round UP) [A1] Accurate (rounded £35.47 (or £35.48))

Alternative Method (Convert to £ first):

- Meal in £ = €38.90 ÷ 1.127 = £34.5164...
- Bank fee = 2.75% of £34.5164... = £0.9492...
- Total in £ = £34.5164... + £0.9492... = £35.4656... ≈ £35.47
- Both methods give identical result of £35.47

£ **35.47**

(Total for Question 11 is 5 marks)

12 Bilal works for a supermarket.
His usual pay is £8.70 per hour.

Bilal will work for $7\frac{1}{2}$ hours this Sunday.
The supermarket will pay Bilal time and a half for working on a Sunday.

Bilal thinks the supermarket will pay him more than £100 for working this Sunday.

(a) Is he correct?
Show why you think this.

(3)

Step 1: Calculate Sunday hourly rate (time and a half)

Time and a half = 1.5 × Usual pay rate

[M1] Rate = £13.05 / hr

Sunday rate = £8.70 × 1.5 = £13.05 per hour

(£8.70 + £4.35 = £13.05)

Step 2: Calculate total pay for 7½ hours (7.5 hours)

Total Sunday pay = 7.5 hours × £13.05

= £97.875

[M1] 7.5 × 13.05 = £97.875

≈ £97.88

Step 3: Compare with Bilal's estimate (more than £100)

£97.88 < £100 (He receives £100 − £97.88 = £2.12 less than £100)

Conclusion: Bilal is NOT correct.

[A1] Decision "No" with £97.88

No, he gets £97.88 (< £100)

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Mina works in a different shop.

The table shows some information about the wages for Bilal and the wages for Mina for the last six weeks.

		1	2	3	4	5	6
Wages (£)	**Bilal**	216.43	283.94	192.87	221.02	205.48	186.32
	Mina	296.56	308.71	188.96	259.32	126.40	283.21

Bilal says,

"My wages were more consistent than Mina's wages for the last six weeks."

(b) Were Bilal's wages more consistent than Mina's wages?
Give a reason for your answer.

(3)

Consistency in statistics is evaluated by comparing the RANGE (spread) of wages.
A SMALLER range represents more consistent wages (less variability).

Step 1: Calculate Bilal's wage range over the 6 weeks

- **Bilal's Highest wage = £283.94 (Week 2)**
- **Bilal's Lowest wage = £186.32 (Week 6)**
- **Bilal's Range = £283.94 − £186.32 = £97.62**

[M1] Bilal range = £97.62

Step 2: Calculate Mina's wage range over the 6 weeks

- **Mina's Highest wage = £308.71 (Week 2)**
- **Mina's Lowest wage = £126.40 (Week 5)**
- **Mina's Range = £308.71 − £126.40 = £182.31**

[M1] Mina range = £182.31

Decision: YES, Bilal's wages were more consistent.

Reason: Bilal's wage range (£97.62) is smaller than Mina's wage range (£182.31).
A smaller range shows that Bilal's weekly pay varied less and was more consistent.
[A1] Valid decision "Yes" with correct comparison of ranges (£97.62 < £182.31)

(Total for Question 12 is 6 marks)

TOTAL FOR SECTION B IS 48 MARKS
TOTAL FOR PAPER IS 64 MARKS

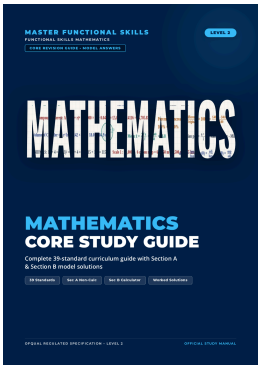
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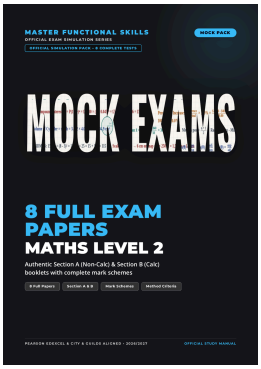


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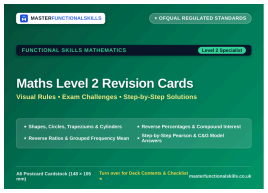


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